

Chapter 1

Black-Scholes Model

1.1 Model Setup: Stock and Bond Dynamics

Call option price formula is shown by

$$C = S_0 \mathbb{N}(d_1) - Ke^{-rT} \mathbb{N}(d_2)$$

Put option price can be calculated by Put-Call Parity

$$P + S_0 = C + Ke^{-rT}$$

Assuming we have a portfolio with underlying stock S and risk-free asset B , differential equations are shown by

$$dB_t = rB_t dt \quad (1.1)$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1.2)$$

where $dW_t = \varepsilon \sqrt{dt}$; $\varepsilon \sim \mathcal{N}(0,1)$, which is a Ito process. μ is drift (deterministic) and dW_t is a diffusion (uncertainty).

Solving S_t and B_t from differential equations

B_t and S_t can be expressed as

$$B_t = B_0 e^{rt} \quad (1.3)$$

Proof:

$$\begin{aligned} \int \frac{dB_t}{B_t} &= \int r dt \\ \int_0^t \ln B_t &= \int_0^t r dt \\ \ln(B_t/B_0) &= r * t \\ B_t &= B_0 e^{rt} \end{aligned}$$

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$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right] \quad (1.4)$$

Proof: Let $Y_t = \ln S_t$

$$\begin{aligned} dY_t &= \frac{\partial Y_t}{\partial t} dt + \frac{\partial Y_t}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 Y_t}{\partial S_t^2} dS_t^2 \\ &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t \end{aligned}$$

we get

$$\begin{aligned} Y_t &= Y_0 + \int_0^t (\mu - \frac{1}{2}\sigma^2)ds + \int_0^t \sigma dW_s \\ &= Y_0 + (\mu - \frac{1}{2})t + \sigma W_t \end{aligned}$$

Given $Y_t = \ln S_t$ Finally, we have to show

$$\begin{aligned} \ln S_t &= \ln S_0 + (\mu - \frac{1}{2})t + \sigma W_t \\ \implies S_t &= S_0 \exp \left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t \right] \end{aligned}$$

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1.2 Risk-Neutral Martingale Pricing Derivation

Proposition 1.2.1

In the Black-Scholes model, the price process Π_t for every traded asset, be it the underlying or derivative asset, has the property that the normalised price process

$$Z_t = \frac{\Pi_t}{B_t}$$

is a martingale under the $\mathbb{Q}$ measure.

In this case,

$$Z_t = \frac{S_t}{B_t}$$

Its differential equation can be written as

$$dZ_t = d\left(\frac{S_t}{B_t}\right)$$

$d\left(\frac{1}{B_t}\right)$ can be calculated by Ito Lamma

$$\begin{aligned} d\left(\frac{1}{B_t}\right) &= 0dt + \left(-\frac{1}{B_t^2}\right)dB_t + \frac{1}{2} \cdot \left(-\frac{2}{B_t^3}\right)(dB_t)^2 \\ &= -\frac{1}{B_t^2} \cdot rB_t dt - 0 \\ &= -\frac{1}{B_t} r dt \end{aligned}$$

Using product rule, then we have

$$\begin{aligned} d\left(\frac{S_t}{B_t}\right) &= \frac{1}{B_t} \cdot dS_t + S_t \cdot d\left(\frac{1}{B_t}\right) + dS_t \cdot d\left(\frac{1}{B_t}\right) \\ &= \frac{1}{B_t} \cdot (\mu S_t dt + \sigma S_t dW_t) + S_t \cdot \left(-\frac{r}{B_t}\right) dt + 0 \\ &= \frac{S_t}{B_t} \mu dt + \frac{S_t}{B_t} \sigma dW_t - \frac{S_t}{B_t} r dt \\ &= \frac{S_t}{B_t} (\mu - r) dt + \frac{S_t}{B_t} \sigma dW_t \end{aligned}$$

Namely under $\mathbb{P}$ measure

$$dZ_t = Z_t(\mu - r)dt + Z_t\sigma dW_t$$

Transfer $\mathbb{P}$ to $\mathbb{Q}$ measure

Theorem 1.2.1 Novikov Condition

θ satisfies the Novikov Condition if $\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \theta_s^2 ds \right) \right] < \infty$

If θ satisfies Novikov condition, the process $\mathcal{M}^\theta$ defined as

$$\mathcal{M}_t^\theta = \exp \left(\int_0^t \theta_s dX_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right), t \in [0, T]$$

is a martingale.

Theorem 1.2.2 Girsanovs Theorem

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(\int_0^t \theta_s dX_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right)$$

In this case, the process $X^\mathbb{Q}$ is defined as

$$X_t^\mathbb{Q} = X_t^\mathbb{P} - \int_0^t \theta_s ds, t \in [0, T]$$

is a standard Brownian Motion.

From Girsanovs Theorem, we have

$$W_t^\mathbb{Q} = W_t^\mathbb{P} + \int_0^t \theta_s ds$$

and

$$dW_t^\mathbb{Q} = dW_t + \theta_t dt$$

$$\begin{aligned} dZ_t &= (\mu - r)Z_t dt + \sigma Z_t dW_t^\mathbb{P} \\ &= (\mu - r)Z_t dt + \sigma Z_t (\theta_t dt + dW_t^\mathbb{Q}) \\ &= (\mu - r + \sigma \theta_t)Z_t dt + \sigma Z_t dW_t^\mathbb{Q} \end{aligned}$$

dZ_t is a martingale under $\mathbb{Q}$ dynamics Therefore

$$\mu - r + \sigma \theta_t = 0$$

θ_t can be expressed as

$$\theta_t = \frac{\mu - r}{\sigma}$$

And θ_t is the Sharpe Ratio. μ is the return of underlying stock S and r is the risk-free rate.

Under $\mathbb{Q}$ measure

$$dZ_t = \sigma Z_t dW_t^\mathbb{Q}$$

Portfolio Replication

Return back to our portfolio. Φ_t^S and Φ_t^B are the weights of the stock and risk-free asset respectively.

The value of the portfolio V_t and its differential equation are shown as

$$V_t = \Phi_t^S S_t + \Phi_t^B B_t$$

$$dV_t = \Phi_t^S dS_t + \Phi_t^B dB_t$$

Checking if V_t is a martingale

$$\begin{aligned}
d\left(\frac{V_t}{B_t}\right) &= dV_t * \frac{1}{B_t} + V_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^S dS_t + \Phi_t^B dB_t * \frac{1}{B_t} + V_t + V_t = \Phi_t^S S_t + \Phi_t^B B_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^S dS_t * \left(\frac{1}{B_t}\right) + \Phi_t^B dB_t * \left(\frac{1}{B_t}\right) + \Phi_t^S S_t * \left(-\frac{1}{B_t}\right) r dt + \Phi_t^B B_t * \left(-\frac{1}{B_t}\right) r dt \\
&= \Phi_t^S \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t \right] + \Phi_t^B \left[\left(\frac{1}{B_t}\right) * r B_t dt - r B_t * \left(\frac{1}{B_t}\right) dt \right] \\
&= \Phi_t^S \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t \right] \\
&= \Phi_t^S \left[S_t * d\left(\frac{1}{B_t}\right) + \left(\frac{1}{B_t}\right) * dS_t \right] \\
&= \Phi_t^S * d\left(\frac{S_t}{B_t}\right) \\
&= \Phi_t^S dZ_t
\end{aligned}$$

Under $\mathbf{Q}$ measure

$$d\left(\frac{V_t}{B_t}\right) = \Phi_t^S dZ_t = \Phi_t^S * \sigma Z_t dW_t^{\mathbf{Q}}$$

Let

$$G_t = \frac{V_t}{B_t}$$

Then G_t is a martingale under $\mathbf{Q}$ measure

Remark the definition of martingale $\forall s < t, \mathbb{E}[X_t | \mathcal{F}_s] = X_s$, X_t is a Martingale.

Therefore

$$\mathbb{E}^{\mathbf{Q}}[G_T | \mathcal{F}_t] = G_t$$

$\chi(t, S_t)$ is the option value. At time T , the option value is $\chi(T, S_T)$, S_T is the underlying price at time T .

$$\begin{aligned}
V_t &= G_t * B_t \\
&= e^{-rt} \mathbb{E}^{\mathbf{Q}}[G_T | \mathcal{F}_t] \\
&= e^{-rt} \mathbb{E}^{\mathbf{Q}}[V_T * e^{rT} | \mathcal{F}_t] \\
&= e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[V_T | \mathcal{F}_t] \\
&= e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\} | \mathcal{F}_t]
\end{aligned}$$

Therefore

$$V_t = e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\} | \mathcal{F}_t]$$

Pricing Call Option

We want to price the value of a call option at $t = 0$

$$V_0 = e^{-rT} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\}]$$

Let's introduce an indicator function

$$V_0 = e^{-rT} \mathbb{E}^{\mathbf{Q}}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}]$$

Under $\mathbf{P}$ measure

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbf{P}}$$

Under $\mathbb{Q}$ measure

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t^{\mathbb{Q}} \\ S_t &= S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma \varepsilon \sqrt{t}} \end{aligned}$$

and

$$\begin{aligned} S_T &= S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \varepsilon \sqrt{T}} \\ V_0 &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}] \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T \geq K\}}] - e^{-rT} \mathbb{E}^{\mathbb{Q}}[K \mathbf{1}_{\{S_T \geq K\}}] \end{aligned}$$

Lemma 1.2.1 Change of Measure Identity

$$\mathbb{E} \left[e^{m + \lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}} \right] = e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a)$$

where $\varepsilon \sim \mathcal{N}(0, 1)$, m, λ, a are constant.

Proof:

$$\begin{aligned} \mathbb{E} \left[e^{m + \lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}} \right] &= \int_{-\infty}^{\infty} e^{m + \lambda x} \mathbf{1}_{\{x \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \int_a^{\infty} e^{m + \lambda x} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \int_a^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\lambda)^2}{2}} e^{\frac{\lambda^2}{2} + m} dx \\ &= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} e^{\frac{\lambda^2}{2} + m} dy \\ &= e^{\frac{\lambda^2}{2} + m} [\mathcal{N}(\infty) - \mathcal{N}(a - \lambda)] \\ &= e^{\frac{\lambda^2}{2} + m} [1 - \mathcal{N}(a - \lambda)] \\ &= e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a) \end{aligned}$$

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$$\begin{aligned} e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T \geq K\}}] &= e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma \varepsilon \sqrt{T} \right] \mathbf{1}_{\{S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma \varepsilon \sqrt{T}] \geq K\}} \right] \\ &= e^{-rT} S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) T \right] \mathbb{E} \left[\exp^{\sigma \varepsilon \sqrt{T}} \mathbf{1}_{\left\{ \varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right\}} \right] \\ &= S_0 \exp \left(-\frac{1}{2}\sigma^2 T \right) \mathbb{E}^{\mathbb{Q}} \left[\exp^{0 + \sigma \sqrt{T} \varepsilon} \mathbf{1}_{\left\{ \varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right\}} \right] \\ &= S_0 \exp \left(-\frac{1}{2}\sigma^2 T \right) * \exp \left(\frac{\sigma^2 T}{2} + 0 \right) * \mathcal{N} \left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 * \mathcal{N} \left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 \mathcal{N} \left(\frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 \mathcal{N}(d_1) \end{aligned}$$

$$\begin{aligned}
e^{-rT} \mathbb{E} [K \mathbf{1}_{\{S_T \geq K\}}] &= K e^{-rT} \mathbb{E} [\mathbf{1}_{\{S_T \geq K\}}] \\
&= K e^{-rT} [1 * Q(S_T \geq K) + 0 * Q(S_T < K)] \\
&= K e^{-rT} * Q(S_T \geq K) \\
&= K e^{-rT} Q\left(\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}\left(\frac{\ln(\frac{S_0}{K}) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}(d_2)
\end{aligned}$$

Therefore, the call option price

$$V_0 = S_0 \mathbf{N}(d_1) - K e^{-rT} \mathbf{N}(d_2)$$

1.3 Delta-Hedging PDE Derivation

Considering we have an option and we want to eliminate the risk by buying some shares of stock.

$$\Pi = V - \Delta S$$

Its differential equation is

$$d\Pi = dV - \Delta dS$$

Remark dS and dV

$$\begin{aligned}
dS_t &= \mu S_t dt + \sigma S_t dW_t \\
dV_t &= \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt + \frac{\partial V}{\partial S} dS
\end{aligned}$$

Plugging dV_t to $d\Pi$, we have

$$d\Pi_t = \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt + \left(\frac{\partial V}{\partial S} - \Delta \right) dS$$

Since

$$\Delta = \frac{\partial V}{\partial S}$$

Then

$$d\Pi_t = \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt$$

Since the hedged portfolio is locally riskless, by no-arbitrage it must earn the risk-free rate. We have

$$d\Pi_t = \Pi r dt$$

Therefore

$$\left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt = r \left(V - S \frac{\partial V}{\partial S} \right) dt$$

Removing right side two terms to left side, we get

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$

Theorem 1.3.1 Feynman-Kac

Assume that F is a solution to the boundary value problem:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$
$$F(T, x) = \Phi(S)$$

Then F has the representation

$$F(t, x) = e^{-r(T-t)} \mathbb{E}[\Phi(S(T))]$$

The call option price is

$$V = e^{-rT} \mathbb{E}[(S_T - K)^+]$$

1.4 Self-Financing Replication Derivation

We set the χ is a option, and $G(S_t)$ is the value at the expiry date.
 V_t is the value of the option at the expiry date.

$$V_t = V_0 + \int_0^t \Phi_u^S dS_u + \Phi_u^B dB_u \quad (1.5)$$

Note:-

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

To derivative Black-Scholes, we have to solutions that kills dW_t or dt .

1. kill dW_t and remain dt , hedge risk (uncertainty) long call and short Δ stock (risk management) $\rightarrow PDE$
2. kill dt and remain dW_t , we get $f(t, S_t)dW_t$, which is a martingale

Set

$$S_t^* = S_t e^{-rt}$$

$$\begin{array}{ccc} | & & | \\ S_t^* & & S_t \end{array} \quad \begin{array}{c} | \\ S_T \end{array}$$

$$d(X_t, Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$$

$$de^{-rt} = -re^{-rt} dt$$

$$dt^2 = 0$$

$$dt dW_t = 0$$

$$dW_t^2 = dt$$

we have

$$\begin{aligned} dS_t^* &= d(S_t e^{-rt}) \\ &= S_t de^{-rt} + e^{-rt} dS_t + dS_t de^{-rt} \\ &= S_t(-e^{-rt} dt) + e^{-rt}(\mu S_t dt + \sigma S_t dW_t) \\ &= (\mu - r)S_t^* dt + \sigma S_t^* dW_t \end{aligned}$$

In general, $\mu - r > 0$ under $\mathbf{P}$ measurement. We can find $\mathbf{Q}$ st. $\{S_t^*, 0 \leq t \leq T\}$ is a $\mathbf{Q}$ -Martingale, where $\mu - r = 0$.
 We need to prove $dS_t^* = \sigma S_t^* dW_t$

The θ_s is unknown, we need to solve it.

$$\begin{aligned} dS_t^* &= (\mu - r)S_t^* dt + \sigma S_t^* dW_t \\ &= (\mu - r)S_t^* dt + \sigma S_t^* (dW_t^{\mathbf{Q}} - \theta_t dt) \\ &= \mu S_t^* dt - r S_t^* dt + \sigma S_t^* dW_t^{\mathbf{Q}} - \sigma S_t^* \theta_t dt \\ &= (\mu - r - \sigma \theta_t)S_t^* dt + \sigma S_t^* dW_t^{\mathbf{Q}} \end{aligned}$$

Let $\mu - r - \theta_t = 0$ to kill drift (dt), under $\theta_t = \frac{(\mu - r)}{\sigma}$ (Sharpe Ratio)

Finally, we get

$$dS_t^* = \sigma S_t^* dW_t^{\mathbf{Q}}$$

and we conclude that S_t^* is a $\mathbf{Q}$ -Martingale.

Definition 1.4.1: Martingale

$\forall s < t \implies \mathbb{E}[X_t | \mathcal{F}_s] = X_s$, X_t is a Martingale.

$\chi(t, S_t)$ is the option value and V_t is the value of the portfolio replicated by risk-free asset and stocks. So, $\chi(t, S_t) = V_t$. We set V_t^* is the discounted value of V_t , where $V_t^* = V_t e^{-rt}$.

$$\begin{aligned}
dV_t^* &= d(V_t e^{-rt}) \\
&= V_t de^{-rt} + e^{-rt} dV_t + dV_t de^{-rt} \\
&= (\Phi_t^S S_t + \Phi_t^B B_t) r e^{-rt} dt + (\Phi_t^S S_t + \Phi_t^B B_t) e^{-rt} d\ln V_t \\
&= \Phi_t^S (-r S_t + e^{-rt} dt + e^{-rt} dS_t) \\
&= \Phi_t^S (S_t de^{-rt} + r^{-rt} dS_t) \\
&= \Phi_t^S d(S_t e^{-rt}) \\
&= \Phi_t^S dS_t^*
\end{aligned}$$

$$dV_t^* = \Phi_t^S dS_t^*$$

Given $dS_t^* = \sigma S_t^* dW_t^Q$
We get

$$dV_t^* = \Phi_t^S \sigma S_t^* dW_t^Q$$

$$\mathbb{E}[V_T | \mathcal{F}_t] = V_t^*$$

$$V_T^* = V_T e^{-rT} = \chi(T, S_T) e^{-rT} = G(S_T) e^{-rT}$$

$$\mathbb{E}[V_T^* | \mathcal{F}_t] = \mathbb{E}[G(S_T) e^{-rT} | \mathcal{F}_t]$$

Claim 1.4.1 Call Option Pricing

$$C = e^{-rT} \mathbb{E}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}]$$

Lemma 1.4.1

$$\mathbb{E} [e^{m+\lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}}] = e^{\frac{\lambda^2}{2} + m} \mathbf{N}(\lambda - a)$$

where $\varepsilon \sim \mathbf{N}(0, 1)$, m, λ, a are constant.

Proof:

$$\begin{aligned}
\mathbb{E} [e^{m+\lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}}] &= \int_{-\infty}^{\infty} e^{m+\lambda x} \mathbf{1}_{\{x \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&= \int_a^{\infty} e^{m+\lambda x} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&= \int_a^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\lambda)^2}{2}} e^{\frac{\lambda^2}{2} + m} dx \\
&= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} e^{\frac{\lambda^2}{2} + m} dy \\
&= e^{\frac{\lambda^2}{2} + m} [\mathcal{N}(\infty) - \mathcal{N}(a - \lambda)] \\
&= e^{\frac{\lambda^2}{2} + m} [1 - \mathcal{N}(a - \lambda)] \\
&= e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a)
\end{aligned}$$

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Proof:

$$\begin{aligned}
e^{-rT} \mathbb{E}^Q[S_T \mathbf{1}_{\{S_T \geq K\}}] &= e^{-rT} \mathbb{E}^Q \left[S_0 \exp\left[\left(r - \frac{1}{2}\right)T + \sigma \varepsilon \sqrt{T}\right] \mathbf{1}_{\{S_0 \exp\left[\left(r - \frac{1}{2}\right)T + \sigma \varepsilon \sqrt{T}\right] \geq K\}} \right] \\
&= e^{-rT} S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T\right] \mathbb{E} \left[\exp^{\sigma \varepsilon \sqrt{T}} \mathbf{1}_{\left\{\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right\}} \right] \\
&= S_0 \exp\left(-\frac{1}{2}\sigma^2 T\right) \mathbb{E}^Q \left[\exp^{0 + \sigma \sqrt{T} \varepsilon} \mathbf{1}_{\left\{\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right\}} \right] \\
&= S_0 \exp\left(-\frac{1}{2}\sigma^2 T\right) * \exp\left(\frac{\sigma^2 T}{2} + 0\right) * \mathbf{N}\left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 * \mathbf{N}\left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 \mathbf{N}\left(\frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 \mathbf{N}(d_1)
\end{aligned}$$

$$\begin{aligned}
e^{-rT} \mathbb{E}[K \mathbf{1}_{\{S_T \geq K\}}] &= K e^{-rT} \mathbb{E}[\mathbf{1}_{\{S_T \geq K\}}] \\
&= K e^{-rT} [1 * Q(S_T \geq K) + 0 * Q(S_T < K)] \\
&= K e^{-rT} * Q(S_T \geq K) \\
&= K e^{-rT} Q\left(\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}\left(\frac{\ln(\frac{S_0}{K}) + (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}(d_2)
\end{aligned}$$

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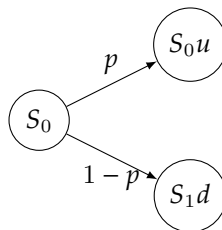
Claim 1.4.2 Call Option Pricing

$$C = S_0 \mathbf{N}(d_1) - K e^{-rT} \mathbf{N}(d_2)$$

1.5 Bridging the Gap: Binomial Tree Delta-Hedging and its Convergence to Black-Scholes

Discrete Binomial Tree

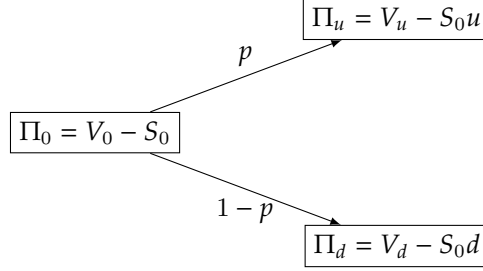
Consider a single period binomial tree



We have a portfolio with a option and stock, V and S are the values respectively. Π is the portfolio value. We want to use delta hedging to elimiate the risk, which is shown by

$$\Pi = V - \Delta S$$

where Δ is the number of the shares.



We want to pick a Δ to make the equation holding

$$\Pi_u = \Pi_d, \quad V_u - S_0 u = V_d - S_0 d$$

The Δ is solved by

$$\Delta = \frac{V_u - V_d}{S_0(u - d)}$$

Assuming maturity is δt and the discounted factor

$$D = \frac{1}{B} = e^{-r\delta t}$$

Under assumption of arbitrage free,

$$\Pi_0 = e^{-r\delta t}[V_u - \Delta S_u] = e^{-r\delta t}[V_d - \Delta S_d]$$

Namely

$$V_0 - \Delta S_0 = e^{-r\delta t}[V_u - \Delta S_0 u] = e^{-r\delta t}[V_d - \Delta S_0 d]$$

V_0 is solved as

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

In risk neutral world, investors must earn the risk-free rate on average. We have

$$\begin{aligned} S_0 e^{r\delta t} &= \mathbb{E}[S_{\delta t}] \\ &= q * S_0 u + (1 - q)S_0 d \end{aligned}$$

The probability of moving up under $\mathbf{Q}$ is

$$q = \frac{e^r \delta t - d}{u - d}$$

The option price V value at $t = 0$ is

$$V_0 = e^{-r\delta t}[q * V_u + (1 - q) * V_d]$$

While the replication argument is based on hedging, we can simplify the formula by introducing a synthetic probability q . We define the "Risk-Neutral World" as a measure $\mathbf{Q}$ where the expected return of the stock is the risk-free rate:

$$S_0 e^{r\delta t} = \mathbb{E}^{\mathbf{Q}}[S_{\delta t}] = qS_0 u + (1 - q)S_0 d$$

Solving for q (the risk-neutral probability of an upward move):

$$q = \frac{e^{r\delta t} - d}{u - d}$$

Under this measure $\mathbf{Q}$, the option price becomes a simple discounted expectation:

$$V_0 = e^{-r\delta t} \mathbb{E}^{\mathbf{Q}}[V_{\delta t}] = e^{-r\delta t}[qV_u + (1 - q)V_d]$$

Therefore, the option price at $t = 0$ can be written as

$$V_0 = \mathbb{E}^Q[e^{-r\delta t} V_T] = e^{-r\delta t} [q * V_u + (1 - q)V_d]$$

Since Cox-Ross-Rubinstein, we have

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = \frac{1}{u}, \quad D = e^{-r\delta t}$$

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

Multiply by $(u - d) * e^{r\delta t}$ both sides

$$(u - d)V_0 * e^{r\delta t} = e^{r\delta t}(V_u - V_d) + (uV_d - dV_u)$$

Using Taylor Expansion, we have

$$u = e^{\sigma\sqrt{\delta t}} \approx 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$$

$$d = \frac{1}{u} \approx 1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$$

$$e^{r\delta t} \approx 1 + r\delta t$$

Continuous Binomial Tree

We follow the CRR (Cox-Ross-Rubinstein) Convergence path.

Step 1: The Multi-period Binomial Formula In a binomial tree with n steps, the option price V_0 is the discounted expected value of all possible payoff paths. Let k be the number of "up" moves:

$$V_0 = e^{-rT} \sum_{k=0}^n \binom{n}{k} q^k (1 - q)^{n-k} \max(S_0 u^k d^{n-k} - K, 0)$$

Let a be the minimum number of up moves required for the option to finish in-the-money ($S_T \geq K$). We split the sum into two parts:

$$V_0 = S_0 \left[\sum_{k=a}^n \binom{n}{k} q^k (1 - q)^{n-k} \frac{u^k d^{n-k}}{e^{rT}} \right] - K e^{-rT} \left[\sum_{k=a}^n \binom{n}{k} q^k (1 - q)^{n-k} \right]$$

Step 2: Parameter Setup (CRR) and Taylor Expansion To ensure the discrete model matches the Geometric Brownian Motion (GBM) as $\delta t \rightarrow 0$, we use:

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}}, \quad q = \frac{e^{r\delta t} - d}{u - d}$$

Using Taylor Expansion $e^x \approx 1 + x + \frac{1}{2}x^2$: $u \approx 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$, $d \approx 1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$, $e^{r\delta t} \approx 1 + r\delta t$. Substituting these into q :

$$q \approx \frac{(1 + r\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}{(1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)} \approx \frac{\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t}{2\sigma\sqrt{\delta t}} = \frac{1}{2} + \frac{1}{2\sigma} \left(r - \frac{1}{2}\sigma^2 \right) \sqrt{\delta t}$$

Step 3: Application of Central Limit Theorem (CLT) As $n \rightarrow \infty$, the binomial distribution $B(n, q)$ converges to a Normal Distribution $N(\mu, \sigma^2)$. The probability of the option being in-the-money (the second bracket in Step 1) is:

$$P(k \geq a) = P\left(\frac{k - nq}{\sqrt{nq(1 - q)}} \geq \frac{a - nq}{\sqrt{nq(1 - q)}} \right)$$

Using the CLT, this probability converges to the cumulative normal distribution:

$$\lim_{n \rightarrow \infty} P(k \geq a) = N(d_2)$$

Where $d_2 = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$. Step 4: The Change of Numeraire for $N(d_1)$ The first bracket in Step 1 involves an extra term $u^k d^{n-k}$. We define a new probability measure q' (often called the stock measure):

$$q' = q \frac{u}{e^{r\delta t}}$$

By substituting q' into the summation, we get a similar normal convergence:

$$\lim_{n \rightarrow \infty} \sum_{k=a}^n \binom{n}{k} (q')^k (1 - q')^{n-k} = \mathbf{N}(d_1)$$

Where $d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$. Final Convergence Result Combining the limits from Step 3 and Step 4 back into the Step 1 formula, we arrive at the Black-Scholes Call Price:

$$C = S_0 \mathbf{N}(d_1) - Ke^{-rT} \mathbf{N}(d_2)$$

1.6 Comparison of the Three Perspectives

The risk-neutral martingale approach derives the option price by changing from the physical probability measure $\mathbf{P}$ to the risk-neutral measure $\mathbf{Q}$. Under $\mathbf{Q}$, discounted asset prices become martingales, and the derivative price can be written as the discounted expected payoff.

The delta-hedging approach derives the Black-Scholes PDE by constructing a locally riskless portfolio. By choosing Δ equal to the derivative of the option value with respect to the stock price, the stochastic risk term is eliminated.

The self-financing replication approach shows that the option payoff can be replicated by dynamically trading the stock and bond. Therefore, by the no-arbitrage principle, the option price must equal the cost of the replicating portfolio.

These three perspectives are closely related. Delta hedging constructs the replicating strategy, replication justifies the no-arbitrage price, and martingale pricing provides the compact expectation formula.

Although the Black-Scholes formula can be derived in many different ways, these derivations are not completely independent. Most of them are different mathematical representations of the same no-arbitrage principle in a complete market. Delta hedging, self-financing replication, and risk-neutral martingale pricing form the three core perspectives. Other methods, such as Feynman–Kac, Girsanov theorem, heat-equation transformation, lognormal integration, change of numeraire, and the binomial limit, can be viewed as refinements or extensions of these core perspectives.